Universality for orthoscheme reflection groups: the right-angled Coxeter envelope and the collision depth

Vico Bonfioli*

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Abstract

The $n+1$ facet reflections of an n-dimensional right-corner orthoscheme $O(a_1,\dots,a_n)$ generate a group $W(a) \le \mathrm{Isom}(\mathbb{R}^n)$, and $W(a)$ is a quotient of the right-angled Coxeter group W_n on the orthogonality graph of the facets. This paper determines what the quotient can do to the growth.

Three statements are proved unconditionally. First, a dimension-free dichotomy: if $Q = G/N$ is a quotient of a marked group and $K(N)$ is the length of a shortest nontrivial element of N, then the sphere sizes of Q and of G agree in every degree below $\lceil K(N)/2 \rceil$ and Q is strictly smaller in that degree (Theorem 4). Second, the growth series of W_n is rational, equal to $(1+t)^{\lfloor n/2 \rfloor +1}/K_{n+1}(t)$ for an explicit integer polynomial K_{n+1}, with growth rate $r_n = 1 + 2\cos\bigl(2\pi/(n+3)\bigr)$ (Theorem 8). Third, a complete classification of the rank-two relations: $\ker(W_n \to W(a))$ meets a rank-two standard parabolic nontrivially if and only if the leg sequence has two equal legs at an endpoint of the staircase or three consecutive equal legs (Theorem 16). The Diophantine input is that three quartics in the flanking legs have only trivial rational points, their Jacobians being the rank-zero curves `24a1`, `72a2` and $y^2 = x^3 - x$.

Combining the first two, the collision depth $\mathrm{cd}_n(a) = \inf\{d : u_d(a) < [t^d]\, W_n(t)\}$ is exactly $K(\ker \rho_a)/2$. We show $K(\ker \rho_a) \ge 6$ for every positive leg tuple, so $\mathrm{cd}_n \ge 3$ always, and we determine cd_n from the two positional statistics (e,t) of the leg sequence: unconditionally when three consecutive legs are equal, and otherwise up to one explicitly stated hypothesis (Theorem 21).

What remains open is whether ρ_a is injective at all for $n \ge 3$; it is open in both directions, and we state it as one problem in two forms (Problem 27). In the plane ρ_a is not injective, by an amenability argument, and that argument does not extend: $O(n)$ contains a free subgroup of rank two for every $n \ge 3$, so $\mathrm{Isom}(\mathbb{R}^n)$ is not amenable as an abstract group (Remark 26). We also record that nonvanishing of the Schur-complement determinant $\det Q_n = -\prod_i a_i^2$ (Lemma 18) does not imply faithfulness, contrary to an argument used in an earlier version of this material (Remark 19).

1. Introduction

An *n-dimensional right-corner orthoscheme*, or path simplex, $O = O(a_1,\dots,a_n) \subset \mathbb{R}^n$ is the simplex with vertices

$$V_0 = 0, \qquad V_k = V_{k-1} + a_k e_k \quad (1 \le k \le n),$$

where $e_1,\dots,e_n$ is the standard basis and $a = (a_1,\dots,a_n)$ is a tuple of positive reals, the *legs*. Its $n+1$ facets carry reflections $\widehat{R}_0,\dots,\widehat{R}_n \in \mathrm{Isom}(\mathbb{R}^n)$, and we write $W(a) = \langle \widehat{R}_0,\dots,\widehat{R}_n \rangle$ for the group they generate.

*Independent researcher, Italy. `vicobonfioli@gmail.com`. https://elvec1o.github.io/home/posts/.

Two facets $\widehat{R}_i, \widehat{R}_j$ with $|i-j| \geq 2$ are perpendicular for every leg tuple (Lemma 1), while consecutive facets meet at an angle that depends on the legs and is generically not a rational multiple of π. The largest group compatible with the leg-independent relations is therefore the right-angled Coxeter group

$$W_n = \langle R_0, \ldots, R_n \;\mid\; R_i^2 = 1,\; (R_i R_j)^2 = 1 \;(|i-j| \geq 2)\rangle, \tag{1}$$

on the *orthogonality graph* Γ_n (vertices $0, \ldots, n$, an edge whenever $|i-j| \geq 2$), and there is a surjection $\rho_a \colon W_n \twoheadrightarrow W(a)$ for every a. Distinguishing W_n from $W(a)$ is essential: writing W_n for the geometric group would make "ρ_a is faithful" vacuous, and it is exactly the possible failure of injectivity that this paper is about.

1.1 Results

The sphere sizes $u_d(a) = \#\{w \in W(a) : \ell(w) = d\}$ are bounded above by the coefficients of the growth series $W_n(t)$ of the envelope (1), and the whole question is where and by how much the bound fails to be sharp. Three unconditional theorems answer the "where".

- **The dichotomy** (Theorem 4). For any group G with a finite symmetric generating set and any $N \trianglelefteq G$ with $N \neq \{1\}$, put $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$. Then the sphere sizes of G/N agree with those of G in every degree $d < \lceil K(N)/2 \rceil$, and are strictly smaller in degree $\lceil K(N)/2 \rceil$. So a family of quotients of a fixed G has a universal initial segment, and the horizon of universality is half the shortest added relator.
- **The envelope** (Theorem 8). $W_n(t)$ is rational, and is $(1+t)^{\lfloor n/2 \rfloor + 1}/K_{n+1}(t)$ for the explicit integer polynomials K_m of (2). Its growth rate is $r_n = 1 + 2\cos(2\pi/(n+3))$.
- **The rank-two classification** (Theorem 16). For $a \in \mathbb{Q}^n_{>0}$, the kernel of ρ_a meets a rank-two standard parabolic $\langle R_i, R_{i+1} \rangle$ nontrivially if and only if $e(a) + t(a) \geq 1$, where $e(a)$ counts equal legs at the two endpoints of the staircase and $t(a)$ counts three consecutive equal legs. The shortest such element has length 6 when $t(a) \geq 1$ and length 8 when $t(a) = 0 < e(a)$.

The first two combine into the statement that gives the paper its shape. Define the *collision depth*

$$\mathrm{cd}_n(a) = \inf\{d \geq 0 : \; u_d(a) < [t^d]\, W_n(t)\} \in \{1, 2, \ldots, \infty\}.$$

By the dichotomy, $\mathrm{cd}_n(a) = \lceil K(\ker \rho_a)/2 \rceil$, and since a relator of a reflection group has even length this is $K(\ker \rho_a)/2$ (Corollary 6). A short geometric argument then shows $K(\ker \rho_a) \geq 6$ for every positive leg tuple (Lemma 7), so $\mathrm{cd}_n \geq 3$ unconditionally, and the rank-two classification pins the value: $\mathrm{cd}_n = 3$ exactly when three consecutive legs are equal, and $\mathrm{cd}_n \in \{3, 4\}$ with the value 4 in every computed case when only an endpoint pair is equal (Theorem 21). The remaining case $e = t = 0$ is $\mathrm{cd}_n = \infty$ precisely when ρ_a is injective.

Whether ρ_a is ever injective for $n \geq 3$ is open in both directions. Section 8 states it as a single problem in two forms, generic and arithmetic, together with the two partial results that bound it: the rank-two half is proved (Theorem 16), and the one soft argument that settles the plane provably does not extend (Remark 26).

1.2 What is and is not new

The dichotomy is elementary. It is recorded here with a proof because it is what turns the two halves of this paper into statements about each other, and because its deviation half needs a case distinction on the parity of $K(N)$ that is easy to omit; but it is not a difficult argument, and the phenomenon it describes, that a family of quotients of a common group shares an initial segment of its growth, is a general one of which the orthoscheme family is an instance. The present paper claims the instance, not the principle.

Both ingredients of r_n are classical and are used as such. Steinberg's formula for the Poincaré series of a Coxeter group is standard [20, 12, 3], growth series of right-angled Coxeter groups are treated systematically by Kellerhals and Perren [14], and the independence polynomials of paths, their Binet formula, and the location of their roots are classical facts about the Fibonacci–Pell family. What is asserted is the identification: that the growth rate of the n-dimensional right-corner orthoscheme envelope is the largest of those roots at $m = n + 3$. We have not located that identification in the literature and make no claim of novelty for the ingredients. A reader who knows the Coxeter growth-series literature should read Theorem 8 as an assembly of known pieces.

The Diophantine part of Theorem 16 rests on three rank-zero Mordell–Weil computations that are in the standard tables [8, 15, 16]; what is done here is the reduction of the dihedral condition to those three curves.

Two points of contact should be recorded. The all-equal tuple $a_1 = \cdots = a_n$ is not a generic point of this family: there the orthoscheme group is the affine Weyl group $\widetilde{C}_n$, whose growth is classical and whose Poincaré series is given by Bott's formula, and which reproduces the $\mathrm{cd}_n = 3$ rows of Table 2. In the planar case $n = 2$, Isola and Piergallini [13] prove that the generic triangle reflection group is not finitely presented, which already implies that ρ_a is not injective there; that is a second reason, besides the amenability argument of Proposition 24, why the plane is not a guide to $n \geq 3$.

1.3 Companion papers

The planar case $n = 2$ is treated separately [2]: there the metric has a lamplighter-type closed form, the deviation from the envelope has been computed past its onset, and the resulting growth series (A396406, growth rate $\beta_2 = 1.4916\ldots$) is transcendental over $\mathbb{Q}(x)$. Nothing below is quoted from that work except the two numerical facts recorded in Corollary 25, and the transcendence questions, which are special to $n = 2$, are not attempted here. Non-right-corner orthoschemes are outside the scope of this paper.

2. Orthoschemes and their Coxeter envelope

Throughout, $n \geq 2$, and

$$U = \{a = (a_1, \ldots, a_n) \in \mathbb{R}^n : \ a_i > 0 \text{ for all } i\}$$

is the leg space. Everything below is invariant under simultaneous scaling of the legs, so one may normalise if convenient; we do not.

2.1 Facet normals

Lemma 1 (Facet normals)**.** *With $V_0 = 0$ and $V_k = \sum_{i \leq k} a_i e_i$, the facet of $O(a)$ opposite V_j has normal*

$$m_0 = e_1, \qquad m_j = a_{j+1} e_j - a_j e_{j+1} \ \ (1 \leq j \leq n-1), \qquad m_n = e_n.$$

Consequently $\mathrm{supp}(m_0) = \{1\}$, $\mathrm{supp}(m_j) = \{j, j+1\}$ *and* $\mathrm{supp}(m_n) = \{n\}$, *so* $m_i \cdot m_j = 0$ *whenever* $|i - j| \geq 2$, *while* $m_i \cdot m_{i+1} \neq 0$ *for every* i *and every* $a \in U$.

Proof. The facet opposite V_j is the affine hull of $\{V_i : i \neq j\}$. For $j = 0$ the difference vectors $V_{k+1} - V_k = a_{k+1} e_{k+1}$, $1 \leq k \leq n-1$, span $\langle e_2, \ldots, e_n \rangle$, whence $m_0 = e_1$. For $j = n$ the vertices $V_0, \ldots, V_{n-1}$ span $\langle e_1, \ldots, e_{n-1} \rangle$, whence $m_n = e_n$. For $1 \leq j \leq n-1$ the difference vectors are $a_k e_k$ for $k \leq j-1$, then $V_{j+1} - V_{j-1} = a_j e_j + a_{j+1} e_{j+1}$, then $a_k e_k$ for $k \geq j+2$. Orthogonality to the first and third families confines m_j to coordinates $\{j, j+1\}$, and orthogonality to the middle vector forces $m_j \propto a_{j+1}e_j - a_je_{j+1}$. Two supports of the listed form are disjoint exactly when $|i-j| \geq 2$. For adjacent indices, $m_0 \cdot m_1 = a_2$, $m_j \cdot m_{j+1} = -a_ja_{j+2}$ for $1 \leq j \leq n-2$, and $m_{n-1} \cdot m_n = -a_{n-1}$, all nonzero because every $a_i > 0$. $\square$

Lemma 1 holds for all legs, with no genericity hypothesis and no case check. Hence $\widehat{R}_i^2 = 1$ always and $\widehat{R}_i\widehat{R}_j = \widehat{R}_j\widehat{R}_i$ whenever $|i-j| \geq 2$, so the assignment $R_i \mapsto \widehat{R}_i$ defines a surjection

$$\rho_a \colon W_n \twoheadrightarrow W(a) \leq \mathrm{Isom}(\mathbb{R}^n)$$

from the right-angled Coxeter group (1) on the orthogonality graph Γ_n [9]. The abstract group W_n does not depend on a.

2.2 Counting conventions

For a group H with a marked finite generating set we write ℓ_H for the word length and $s_d(H) = \#\{h \in H : \ell_H(h) = d\}$ for the sphere sizes. We abbreviate

$$u_d(a) = s_d(W(a)), \qquad W_n(t) = \sum_{d \geq 0} s_d(W_n)\, t^d,$$

the generators being $\widehat{R}_0, \dots, \widehat{R}_n$ and $R_0, \dots, R_n$ respectively. Since $O(a)$ is a fundamental domain for the action of $W(a)$, $u_d(a)$ is also the number of distinct chambers $g \cdot O(a)$ at word distance d, which is what the breadth-first search of §10 enumerates. Being the sphere sizes of a quotient of W_n, they satisfy $u_d(a) \leq [t^d]\, W_n(t)$ for every d.

Finally, set

$$K(a) = \min\{\ell_{W_n}(w) : w \in \ker \rho_a \setminus \{1\}\} \in \{1, 2, \dots\} \cup \{\infty\},$$

with $K(a) = \infty$ when ρ_a is injective.

2.3 Generic rigidity

Before computing anything we record that "the generic orthoscheme group" is a well-defined object: it does not depend on which generic shape one picks. Write $N_{\mathrm{gen}} = \bigcap_{a \in U} \ker \rho_a \trianglelefteq W_n$.

Theorem 2 (Generic rigidity)**.** *For every $n \geq 2$ there is a co-null subset $U_{\mathrm{gen}} \subseteq U$, the complement of a countable union of proper real-algebraic subvarieties, such that $\ker \rho_a = N_{\mathrm{gen}}$, hence $W(a) \cong W_n/N_{\mathrm{gen}}$, for every $a \in U_{\mathrm{gen}}$. Consequently $u_d(a)$ is constant on U_{gen}, equal to $s_d(W_n/N_{\mathrm{gen}})$, and this generic value is the maximum of u_d over all of U. Moreover U_{gen} contains every a whose coordinates are algebraically independent over $\mathbb{Q}$; in particular it is dense.*

Proof. Each facet normal $m_j(a)$ is a fixed polynomial vector in a by Lemma 1, so the reflection $\widehat{R}_j = I - 2\, m_j m_j^{\top}/(m_j \cdot m_j)$, together with its translation part, has entries in $\mathbb{Q}(a)$ with denominator the nowhere-vanishing polynomial $m_j \cdot m_j$. Hence for each $w \in W_n$, choosing any word representative, every entry of $\rho_a(w)$ lies in $\mathbb{Q}(a)$, independently of the representative because ρ_a is well defined. The locus $V_w = \{a \in U : \rho_a(w) = I\}$ is the simultaneous vanishing of finitely many rational functions cleared of their nonvanishing denominators, hence Zariski-closed in the irreducible variety U. If $w \notin N_{\mathrm{gen}}$ then $\rho_a(w) \neq I$ for some a, so V_w is a proper closed subvariety, of measure zero. As W_n is countable, $B := \bigcup_{w \in W_n \setminus N_{\mathrm{gen}}} V_w$ is a countable union of proper subvarieties, and $U_{\mathrm{gen}} := U \setminus B$ is co-null and dense.

For $a \in U_{\mathrm{gen}}$ one has $N_{\mathrm{gen}} \subseteq \ker \rho_a$ by definition, and $\ker \rho_a \subseteq N_{\mathrm{gen}}$ because any $w \in \ker \rho_a \setminus N_{\mathrm{gen}}$ would place a in $V_w \subseteq B$. Isomorphic groups with marked generators have identical sphere sizes, giving the constant value; for arbitrary a, $\ker \rho_a \supseteq N_{\mathrm{gen}}$ makes $W(a)$ a quotient of W_n/N_{gen}, so $u_d(a) \leq s_d(W_n/N_{\mathrm{gen}})$. Finally each V_w is defined over $\mathbb{Q}$; if the coordinates of a are algebraically independent over $\mathbb{Q}$ then a lies on no proper $\mathbb{Q}$-subvariety, so $a \notin B$. Such a exist since $\dim U = n \geq 2$. $\square$

Corollary 3 (One faithful shape suffices)**.** *If ρ_a is injective for a single $a \in U$, then $N_{\mathrm{gen}} = \{1\}$ and ρ_b is injective for every $b \in U_{\mathrm{gen}}$. Conversely, if ρ_b fails to be injective for some $b \in U_{\mathrm{gen}}$, then ρ_a fails to be injective for every $a \in U$.*

Proof. $N_{\mathrm{gen}} \subseteq \ker \rho_a = \{1\}$, and $\ker \rho_b = N_{\mathrm{gen}}$ for $b \in U_{\mathrm{gen}}$ by Theorem 2. The converse is the contrapositive, using $N_{\mathrm{gen}} \subseteq \ker \rho_a$ for every a. $\square$

Corollary 3 is what allows the generic question and the arithmetic question of §8 to be treated as two forms of one problem in one direction. It gives nothing in the other direction: a specific rational tuple need not lie in U_{gen}, and U_{gen} contains no tuple with algebraically dependent coordinates, hence no rational tuple at all.

3. The universality–deviation dichotomy

Theorem 4 (Universality–deviation dichotomy)**.** *Let $G = \langle S \rangle$ be a group with finite symmetric generating set S, let $N \trianglelefteq G$ with $N \neq \{1\}$, write $Q = G/N$ and $K = K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$, and put $h = \lceil K/2 \rceil$. Then $s_d(Q) = s_d(G)$ for every $d < h$, and $s_h(Q) < s_h(G)$.*

Proof. Write $\pi : G \to Q$.

Agreement below h. Let $d < h$. If $d \geq 1$ then $2d \leq 2h - 2 \leq K - 1 < K$, since $2h \leq K + 1$. Suppose $g, g' \in B_d(G)$ satisfy $\pi g = \pi g'$. Then $g(g')^{-1} \in N$ and $\ell_G(g(g')^{-1}) \leq 2d < K$, so $g = g'$; thus π is injective on $B_d(G)$. It is also length-preserving there: if $\ell_G(g) = d$ and $\ell_Q(\pi g) = e < d$, pick g' with $\ell_G(g') = e$ and $\pi g' = \pi g$; then $\ell_G(g(g')^{-1}) \leq d + e < 2d < K$ forces $g = g'$, contradicting $e < d$. Finally π maps the sphere of radius d in G onto the sphere of radius d in Q, because $\ell_Q(q) = \min\{\ell_G(g) : \pi g = q\}$ realises every q with $\ell_Q(q) = d$ by some g of length exactly d. Hence $s_d(Q) = s_d(G)$.

Strict drop at h. Choose $g_0 \in N \setminus \{1\}$ with $\ell_G(g_0) = K$ and split a geodesic word for it as $g_0 = w_1 w_2$ with $\ell_G(w_1) = h$ and $\ell_G(w_2) = K - h = \lfloor K/2 \rfloor$; both factors are geodesic as written. From $w_1 w_2 \in N$ we get $\pi(w_1) = \pi(w_2^{-1})$, and $w_1 \neq w_2^{-1}$ since otherwise $g_0 = 1$. Surjectivity of π from the radius-h sphere of G onto that of Q, established in the previous paragraph for $d < h$, holds verbatim for $d = h$.

If K is even then $h = K/2 = \lfloor K/2 \rfloor$, so w_1 and w_2^{-1} are two distinct elements of the sphere of radius h in G with the same image, and $s_h(Q) \leq s_h(G) - 1$.

If K is odd then $h = (K+1)/2$ and $\ell_G(w_2^{-1}) = h - 1$, so $\ell_Q(\pi w_1) \leq h - 1 < h$ and the image of w_1 does not lie on the sphere of radius h in Q. That sphere is therefore covered by π applied to the radius-h sphere of G with w_1 removed, and again $s_h(Q) \leq s_h(G) - 1$. $\square$

The parity split at the end is not cosmetic. When K is odd the two halves of a shortest relator lie on spheres of different radii, so no two points of the upstairs sphere need collide, and the count drops for the other reason. Both cases are machine-checked (§10).

Applied to ρ_a, Theorem 4 gives an exact alternative with no genericity and no dimension hypothesis.

Corollary 5 (The envelope alternative)**.** *Exactly one of the following holds.*

- *(i) ρ_a is injective. Then $W(a) \cong W_n$ and $u_d(a) = [t^d]\, W_n(t)$ for every d; in particular the growth series of $W(a)$ is the rational function of Theorem 8, with growth rate r_n.*
- *(ii) ρ_a is not injective. Then $K(a) < \infty$, $u_d(a) = [t^d]\, W_n(t)$ for every $d < \lceil K(a)/2 \rceil$, and $u_d(a) < [t^d]\, W_n(t)$ at $d = \lceil K(a)/2 \rceil$.*

Proof. If $\ker \rho_a = \{1\}$ then ρ_a is an isomorphism of marked groups and (i) is immediate. Otherwise $K(a)$ is finite and Theorem 4 with $G = W_n$, $N = \ker \rho_a$ gives (ii). $\square$

Corollary 6 (The collision depth is half the shortest relator)**.** *For every $a \in U$, $\operatorname{cd}_n(a) = \lceil K(a)/2 \rceil$, and $K(a)$ is even, so $\operatorname{cd}_n(a) = K(a)/2$.*

Proof. The first equality is Corollary 5: below $\lceil K(a)/2 \rceil$ the two sequences agree and at $\lceil K(a)/2 \rceil$ they differ. For parity, let $\varepsilon : W_n \to \{\pm 1\}$ be the sign character, $\varepsilon(R_i) = -1$, which is well defined because all defining relators of (1) have even length; then $\varepsilon(w) = (-1)^{\ell(w)}$. On the other side, $\det \circ \rho_a$ is a homomorphism $W_n \to \{\pm 1\}$ sending each R_i to -1, so $\det \rho_a(w) = \varepsilon(w)$. If $w \in \ker \rho_a$ then $\det \rho_a(w) = 1$, hence $\ell(w)$ is even. $\square$

Lemma 7 (No relator of length four)**.** *For every $n \geq 2$ and every $a \in U$, $K(a) \geq 6$. Consequently $\operatorname{cd}_n(a) \geq 3$.*

Proof. By Corollary 6, $K(a)$ is even, so it suffices to exclude $K(a) \in \{2, 4\}$.

Write H_i for the facet hyperplane carrying $\widehat{R}_i$. The H_i are the $n+1$ facet hyperplanes of a nondegenerate simplex, so they are pairwise distinct and pairwise non-parallel, and $H_i \cap H_j$ is the affine hull of $\{V_m : m \neq i, j\}$, of dimension $n-2$. Distinct index pairs give distinct intersections, because an $(n-2)$-flat contains exactly $n-1$ of the $n+1$ affinely independent vertices.

$K(a) = 2$ would give a geodesic word $R_i R_j$ with $i \neq j$ and $\widehat{R}_i = \widehat{R}_j$, impossible since $H_i \neq H_j$.

Suppose $K(a) = 4$, realised by a geodesic word $R_i R_j R_k R_l$ with $i \neq j$, $j \neq k$, $k \neq l$ and $\widehat{R}_i \widehat{R}_j \widehat{R}_k \widehat{R}_l = 1$, that is $\widehat{R}_i \widehat{R}_j = \widehat{R}_l \widehat{R}_k$. For $H_i \neq H_j$ non-parallel, the isometry $\widehat{R}_i \widehat{R}_j$ is a rotation about $H_i \cap H_j$ through twice the angle between the hyperplanes, an angle in $(0, 2\pi)$; its fixed-point set is therefore exactly $H_i \cap H_j$. The same applies to $\widehat{R}_l \widehat{R}_k$, so $H_i \cap H_j = H_k \cap H_l$ and hence $\{i, j\} = \{k, l\}$ by the previous paragraph. Since $j \neq k$ this forces $i = k$ and $j = l$, so the relation reads $\widehat{R}_i \widehat{R}_j = \widehat{R}_j \widehat{R}_i$, that is $H_i \perp H_j$. For $|i - j| \geq 2$ that relation already holds in W_n and the word $R_i R_j R_i R_j$ is not geodesic; for $|i-j| = 1$ perpendicularity contradicts $m_i \cdot m_{i+1} \neq 0$ in Lemma 1. Both branches are impossible, so $K(a) \neq 4$. $\square$

4. The envelope in closed form

Theorem 8 (The envelope is rational)**.** *Let $c_{\Gamma_n}(x) = \sum_{C \text{ clique}} x^{|C|}$ be the clique polynomial of Γ_n. The growth series of W_n is rational and satisfies*

$$\frac{1}{W_n(t)} = c_{\Gamma_n}\Big(\frac{-t}{1+t}\Big).$$

Equivalently $W_n(t) = (1+t)^{n+1}/J_{n+1}(t)$, where

$$J_m = (1+t)\big(J_{m-1} - t\, J_{m-2}\big), \qquad J_0 = J_1 = 1.$$

One has $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ with

$$K_0 = K_1 = 1, \qquad K_{2j} = K_{2j-1} - tK_{2j-2}, \qquad K_{2j+1} = (1+t)K_{2j} - tK_{2j-1}, \tag{2}$$

so that

$$W_n(t) = \frac{(1+t)^{\lfloor n/2 \rfloor + 1}}{K_{n+1}(t)}.$$

The reciprocals of the roots of K_{n+1} in $(1/3, 1)$ are the numbers $1 + 2\cos\frac{2k\pi}{n+3}$, and the growth rate is the largest of them,

$$r_n = 1 + 2\cos\frac{2\pi}{n+3}.$$

Proof. W_n is a right-angled Coxeter group, so its finite standard parabolics are exactly the cliques of Γ_n, each a group $(\mathbb{Z}/2)^{|C|}$ with growth polynomial $(1+t)^{|C|}$ [9]. Steinberg's formula [20, 3, 12] gives

$$\frac{1}{W_n(t^{-1})} = \sum_{C \text{ clique}} \frac{(-1)^{|C|}}{(1+t)^{|C|}} = c_{\Gamma_n}\Big(\frac{-1}{1+t}\Big).$$

Substituting $t \mapsto t^{-1}$ and simplifying $-1/(1+t^{-1}) = -t/(1+t)$ yields the displayed identity for $1/W_n(t)$. The substitution must be carried inside the clique polynomial; retaining the argument $-1/(1+t)$ gives a different rational function, already at $n=2$.

The cliques of Γ_n are the independent sets of its complement, the path P_{n+1}, so $c_{\Gamma_n} = I_{n+1}$, the independence polynomial of the path on $n+1$ vertices, which obeys $I_m = I_{m-1} + xI_{m-2}$, $I_0 = 1$, $I_1 = 1+x$, and has the coefficient form $I_m(x) = \sum_k \binom{m+1-k}{k} x^k$. Put $J_m(t) = (1+t)^m I_m(-t/(1+t))$. Multiplying the recurrence by $(1+t)^m$ gives $J_m = (1+t)J_{m-1} - t(1+t)J_{m-2}$, which is the stated recurrence, with $J_0 = 1$ and $J_1 = (1+t)(1-t/(1+t)) = 1$. By construction $1/W_n(t) = I_{n+1}(-t/(1+t)) = J_{n+1}(t)/(1+t)^{n+1}$. The factorisation $J_m = (1+t)^{\lfloor m/2 \rfloor} K_m$ and the two-step recurrences (2) follow by induction on m, splitting on the parity of m; cancelling $(1+t)^{\lfloor (n+1)/2 \rfloor}$ against $(1+t)^{n+1}$ leaves the exponent $\lceil (n+1)/2 \rceil = \lfloor n/2 \rfloor + 1$.

For the roots, fix $t > 0$ and solve the linear recurrence. Its characteristic equation is $\lambda^2 - (1+t)\lambda + t(1+t) = 0$, of discriminant $(1+t)(1-3t)$.

If $0 < t < 1/3$ both roots are real with $0 < \lambda_- < \lambda_+ < 1$, their sum being $1+t \le 4/3$ and their product $t(1+t) < 4/9$. Writing $J_m = \alpha\lambda_+^m + \beta\lambda_-^m$, the initial conditions give $\alpha = (1-\lambda_-)/(\lambda_+ - \lambda_-) > 0$ and $\beta = (\lambda_+ - 1)/(\lambda_+ - \lambda_-) < 0$ with $\alpha + \beta = 1$. Then $J_m = \lambda_-^m(\alpha(\lambda_+/\lambda_-)^m + \beta)$, and the bracket is increasing in m with value 1 at $m = 0$, so $J_m > 0$. At $t = 1/3$ the root is double and $J_m = (1+m/2)(2/3)^m > 0$. Hence J_m has no root in $(0, 1/3]$.

If $t > 1/3$ the roots are $\lambda_\pm = \varrho e^{\pm i\theta}$ with $\varrho = \sqrt{t(1+t)}$ and $\cos\theta = \sqrt{(1+t)/(4t)}$, so θ ranges over $(0, \pi/3)$ and is strictly increasing in t, because $(1+t)/(4t) = \frac{1}{4} + \frac{1}{4t}$ is strictly decreasing. Then $J_m = \varrho^m(\cos m\theta - c\sin m\theta)$, where $J_0 = 1$ is automatic and $J_1 = 1$ forces $c\varrho\sin\theta = (t-1)/2$, that is $c = (t-1)/\sqrt{(1+t)(3t-1)}$. Since $\sin^2\theta = (3t-1)/(4t)$, a direct computation gives $\cot 2\theta = (1-t)/\sqrt{(1+t)(3t-1)} = -c$. Therefore

$$J_m = 0 \iff \cot m\theta = c = -\cot 2\theta \iff \frac{\sin((m+2)\theta)}{\sin m\theta \, \sin 2\theta} = 0 \iff (m+2)\theta \in \pi\mathbb{Z}.$$

With $m = n+1$ the roots are exactly $\theta = k\pi/(n+3)$, $1 \le k < (n+3)/3$. Inverting $\cos^2\theta = (1+t)/(4t)$ gives $t = 1/(4\cos^2\theta - 1) = 1/(1 + 2\cos 2\theta)$, so the corresponding t-values are $1/(1 + 2\cos\frac{2k\pi}{n+3})$. As θ increases with t, the smallest positive root is $k = 1$; it exceeds $1/3$ because $1 + 2\cos\frac{2\pi}{n+3} < 3$, and $1 + t > 0$ there, so it is a root of K_{n+1} and not of the stripped factor. Hence $r_n = 1 + 2\cos\frac{2\pi}{n+3}$. □

Remark 9 (Not a path spectral radius). The number $2\cos\frac{2\pi}{n+3}$ is not the spectral radius of the path P_{n+1}, which is $2\cos\frac{\pi}{n+2}$; the two disagree for every $n \ge 2$ (at $n = 2$ they are $0.618\ldots$ and $1.414\ldots$). The angles produced by the proof are $k\pi/(n+3)$, and the growth rate uses $\cos 2\theta$ at $\theta = \pi/(n+3)$, not $\cos\theta$.

Remark 10 (The clique polynomial is a Fibonacci–Pell polynomial). The coefficient form $I_{n+1}(x) = \sum_k \binom{n+2-k}{k} x^k$ used above is the statement that the clique count of Γ_n in size k is $\binom{n+2-k}{k}$, and it identifies c_{Γ_n} with the Fibonacci–Pell polynomial P_{n+3}, where $P_0 = 0$, $P_1 = 1$ and $P_m = P_{m-1} + yP_{m-2}$: the two families satisfy the same three-term recursion, by Pascal's identity $\binom{n-k+2}{k} = \binom{n-k+1}{k} + \binom{n-k+1}{k-1}$, and agree at the bottom. This gives a second reading of the roots. The characteristic equation $x^2 = x + y$ has roots $\alpha, \beta = (1 \pm \sqrt{1+4y})/2$, the Binet formula yields $P_m(y) = (\alpha^m - \beta^m)/(\alpha - \beta)$, so $P_m(y) = 0$ iff $(\alpha/\beta)^m = 1$; setting $\alpha/\beta = e^{2\pi i j/m}$ gives $y_j = -1/(4\cos^2(\pi j/m))$, and translating back through $y = -1/(1+t)$ with

$4\cos^2\vartheta - 1 = 1 + 2\cos 2\vartheta$ returns the same values $1 + 2\cos\frac{2\pi j}{n+3}$. The Pascal step, the Binet formula and the translation identity are machine-checked (§10); the induction that assembles the Pascal step into the identification $c_{\Gamma_n} = P_{n+3}$ is a paper proof.

Corollary 11 (Asymptotic regime)**.** *$r_n \to 3$ as $n \to \infty$, with*

$$3 - r_n = 4\sin^2\Big(\frac{\pi}{n+3}\Big) \sim \frac{4\pi^2}{(n+3)^2}.$$

Since $2\cos(2\pi/m)$ has degree $\phi(m)/2$ over $\mathbb{Q}$, where ϕ denotes Euler's totient, the rate r_n is rational or a quadratic surd exactly when $\phi(n+3) \leq 4$, that is $n+3 \in \{5, 6, 8, 10, 12\}$; see Table 1, where φ denotes the golden ratio.

n	$W_n(t)$	r_n	algebraic form
2	$(1+t)^2/(1-t-t^2)$	$1.6180\ldots$	$\varphi = (1+\sqrt{5})/2$ (golden)
3	$(1+t)^2/(1-2t)$	2.0000	exactly 2
4	$(1+t)^3/(1-2t-t^2+t^3)$	$2.2470\ldots$	$1+2\cos(2\pi/7)$, cubic
5	$(1+t)^3/(1-3t+t^2+t^3)$	$2.4142\ldots$	$1+\sqrt{2}$ (silver)
6	$(1+t)^4/(1-3t+3t^3)$	$2.5321\ldots$	$1+2\cos(2\pi/9)$, cubic
7	$(1+t)^4/(1-4t+3t^2+2t^3-t^4)$	$2.6180\ldots$	$\varphi^2 = (3+\sqrt{5})/2$
8	$(1+t)^5/(1-4t+2t^2+5t^3-2t^4-t^5)$	$2.6825\ldots$	$1+2\cos(2\pi/11)$, degree 5
9	$(1+t)^5/(1-5t+6t^2+2t^3-4t^4)$	$2.7321\ldots$	$1+\sqrt{3}$
10	$(1+t)^6/(1-5t+5t^2+6t^3-7t^4-2t^5+t^6)$	$2.7709\ldots$	$1+2\cos(2\pi/13)$, degree 6

Table 1: The envelope $W_n(t) = (1+t)^{\lfloor n/2\rfloor+1}/K_{n+1}(t)$ and its growth rate $r_n = 1+2\cos(2\pi/(n+3))$. The recurrence (2), its agreement with the Steinberg expansion, the exactness of the division by $(1+t)^{\lfloor (n+1)/2\rfloor}$, and the identification of the smallest positive root of K_{n+1} with $1/r_n$ are certified for $2 \leq n \leq 30$ by the program of §10.

Remark 12 (The orthoscheme sequences in the OEIS)*.* For pairwise distinct legs the layer counts $u_d(a)$ agree with the envelope as far as they have been computed, and the resulting sequences are recorded in the On-Line Encyclopedia of Integer Sequences: A397439 ($n=3$), A397437 ($n=4$), A396927 ($n=5$) and A397438 ($n=6$), with the $n=7$ entry in preparation. Each of these is the coefficient sequence of the rational function $W_n(t)$ of Table 1, with dominant pole $1/r_n$. The planar entry A396406 is of a different nature: it records the deviated planar sequence, which agrees with $W_2(t)$ only up to degree 9 and whose generating function is transcendental over $\mathbb{Q}(x)$ [2]; see Corollary 25.

5. Rank-two relations

This section determines exactly when $\ker\rho_a$ meets a rank-two standard parabolic $\langle R_i, R_{i+1}\rangle$. Throughout it, the legs are positive rationals. By Lemma 1 the only pairs that can carry a nontrivial dihedral relation are the consecutive ones, and their cosines are

$$\cos\theta_{0,1} = \frac{a_2}{\sqrt{a_1^2+a_2^2}}, \qquad \cos\theta_{n-1,n} = \frac{a_{n-1}}{\sqrt{a_{n-1}^2+a_n^2}}, \qquad \cos\theta_{i,i+1} = \frac{a_i a_{i+2}}{\sqrt{(a_i^2+a_{i+1}^2)(a_{i+1}^2+a_{i+2}^2)}} \tag{3}$$

for $1 \leq i \leq n-2$. In each case $\cos^2\theta \in \mathbb{Q}$. By the extended Niven theorem, a consequence of Conway–Jones [6] (if $\theta \in \mathbb{Q}\pi$ and $\cos^2\theta \in \mathbb{Q}$ then $\cos\theta \in \{0, \pm\frac{1}{2}, \pm\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{3}}{2}, \pm 1\}$), a relation $(R_i R_{i+1})^m = 1$ in $W(a)$ forces

$$\cos^2\theta_{i,i+1} \in \{0, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}, 1\}.$$

5.1 The two positional statistics

For $a \in \mathbb{Q}^n_{>0}$ set

$$e(a) = \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0,1,2\},$$
$$t(a) = \#\{i : 1 \le i \le n-2,\ a_i = a_{i+1} = a_{i+2}\}.$$

Thus e counts equal pairs at the two endpoint positions of the staircase and t counts interior triples of three consecutive equal legs. An equal pair sitting strictly inside the staircase, such as $a_2 = a_3$ in $(1,2,2,3)$ at $n=4$, contributes to neither.

5.2 Boundary pairs

Lemma 13 (Boundary pairs)**.** *A boundary pair carries an admissible value of* $\cos^2\theta$ *if and only if the two legs involved are equal, in which case* $\cos^2\theta = \frac{1}{2}$ *and* $\theta = \pi/4$.

Proof. Take the pair $(0,1)$, so $\cos^2\theta = a_2^2/(a_1^2 + a_2^2)$ by (3); the pair $(n-1,n)$ is symmetric. The value 0 forces $a_2 = 0$ and the value 1 forces $a_1 = 0$, both excluded by positivity. The value $\frac{1}{4}$ gives $4a_2^2 = a_1^2 + a_2^2$, that is $(a_1/a_2)^2 = 3$, and the value $\frac{3}{4}$ gives $(a_2/a_1)^2 = 3$; neither is possible over $\mathbb{Q}$. The value $\frac{1}{2}$ gives $a_1^2 = a_2^2$, that is $a_1 = a_2$, and conversely $a_1 = a_2$ gives $\cos\theta = 1/\sqrt{2}$. □

All five cases of Lemma 13 are formalised in Lean 4 (§10).

5.3 Interior pairs: three Diophantine quartics

At an interior pair write $x = a_i$, $y = a_{i+1}$, $z = a_{i+2}$, so that by (3)

$$\cos^2\theta_{i,i+1} = \frac{x^2 z^2}{(x^2+y^2)(y^2+z^2)}.$$

Requiring this to equal a constant c and setting $u = y^2$ gives the quadratic

$$c\,u^2 + c\,(x^2+z^2)\,u + (c-1)\,x^2 z^2 = 0, \tag{4}$$

which has a rational root only if its discriminant $\Delta = c^2(x^2+z^2)^2 - 4c(c-1)x^2z^2$ is the square of a rational. Substituting the three interior values of c and clearing the constant factors gives, in the two flanking legs $x = a_i$ and $z = a_{i+2}$, the Diophantine *quartics*

$$\mathcal{V}_{1/4}:\ x^4+14x^2z^2+z^4 = k^2, \qquad \mathcal{V}_{1/2}:\ x^4+6x^2z^2+z^4 = k^2, \qquad \mathcal{V}_{3/4}:\ 9x^4+30x^2z^2+9z^4 = k^2.$$

It matters that these are quartics in the legs and not conics in $X = x^2$, $Z = z^2$. Written in X and Z the three equations define conics with infinitely many rational points, among them $(X,Z) = (3,2)$ on $X^2 + 6XZ + Z^2 = k^2$ with $k = 7$, and $(X,Z) = (2,5)$ and $(5,9)$ on the other two. Those points are irrelevant here precisely because X and Z must themselves be squares of legs.

Lemma 14. *Each of* $\mathcal{V}_{1/4}$, $\mathcal{V}_{1/2}$, $\mathcal{V}_{3/4}$ *has only trivial solutions over* $\mathbb{Q}^2_{>0}$ *in the legs* (x,z)*: every rational point has* $xz = 0$ *or* $x = z$.

Proof. $\mathcal{V}_{1/2}$. The quartic $x^4 + 6x^2z^2 + z^4 = k^2$ is birationally equivalent to the elliptic curve $E_{1/2} : y^2 = x^3 - x$ (Mordell [18], §17). $E_{1/2}$ has Mordell–Weil rank 0 over $\mathbb{Q}$ by Fermat's classical descent (Cassels [5], Ch. II), and its full rational-point set is the 2-torsion $\{\infty, (0,0), (\pm1, 0)\}$, pulling back to $xz = 0$ or $x = z$.

$\mathcal{V}_{1/4}$. The Jacobian elliptic curve is $E_{1/4} : y^2 = x(x-12)(x-16)$, which is Cremona `24a1` (conductor 24, $j = 35\,152/9$, Mordell–Weil rank 0 over $\mathbb{Q}$, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); see [8] and the LMFDB record [15]. Each of its eight rational points pulls back to $xz = 0$ or $x = z$.

$\mathcal{V}_{3/4}$. The quartic admits a standard conic-to-Weierstrass substitution and reduces to $E_{3/4} : y^2 = x(x-300)(x-1200)$, Cremona `72a2` (conductor 72, the same j-invariant $35\,152/9$ as $E_{1/4}$, of which it is a quadratic twist, also rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); see [8, 16]. Each torsion point again pulls back to a trivial quartic solution. □

Lemma 14 says nothing about the diagonal $x = z$, which is where the interior analysis has to be completed separately.

Lemma 15 (Interior pairs with equal flanking legs)**.** *Suppose $a_i = a_{i+2}$. Then the interior pair $(i, i+1)$ carries an admissible value of $\cos^2\theta$ if and only if $a_i = a_{i+1} = a_{i+2}$, in which case $\cos^2\theta = \frac{1}{4}$ and $\theta = \pi/3$.*

Proof. With $x = z$ the cosine is $\cos\theta = x^2/(x^2+y^2)$, a positive rational. The values 0 and 1 of $\cos^2\theta$ are excluded by positivity. The value $\frac{1}{2}$ requires $\cos\theta = 1/\sqrt{2}$ and the value $\frac{3}{4}$ requires $\cos\theta = \sqrt{3}/2$, neither rational. The value $\frac{1}{4}$ requires $\cos\theta = \frac{1}{2}$, that is $2x^2 = x^2 + y^2$, hence $y = x$; and conversely $x = y = z$ gives $\cos\theta = x^2/(2x^2) = \frac{1}{2}$. □

5.4 The classification

Theorem 16 (Rank-two classification)**.** *Let $n \geq 2$ and $a \in \mathbb{Q}^n_{>0}$. Then $\ker\rho_a$ meets some rank-two standard parabolic $\langle R_i, R_{i+1}\rangle$ nontrivially if and only if $e(a) + t(a) \geq 1$. More precisely:*

(i) if $t(a) \geq 1$, the interior pair at a triple has $\theta = \pi/3$ and $(R_iR_{i+1})^3 \in \ker\rho_a$ is an element of length 6;

(ii) if $t(a) = 0$ and $e(a) \geq 1$, the corresponding boundary pair has $\theta = \pi/4$ and $(R_iR_{i+1})^4 \in \ker\rho_a$ is an element of length 8, and no shorter element of any rank-two standard parabolic lies in $\ker\rho_a$;

(iii) if $e(a) = t(a) = 0$, no rank-two standard parabolic meets $\ker\rho_a$ nontrivially.

Proof. Since $\langle R_i, R_{i+1}\rangle$ is an infinite dihedral standard parabolic of W_n, a word $(R_iR_{i+1})^m$ is geodesic of length $2m$, and $\ker\rho_a$ meets that parabolic nontrivially exactly when $\theta_{i,i+1} \in \pi\mathbb{Q}$, in which case the shortest element is $(R_iR_{i+1})^m$ with $\theta_{i,i+1} = \pi r/m$ in lowest terms. Non-consecutive pairs generate finite parabolics of W_n itself and contribute nothing.

Suppose $\theta_{i,i+1} \in \pi\mathbb{Q}$ for some i. By the extended Niven theorem $\cos^2\theta_{i,i+1} \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. If the pair is a boundary pair, Lemma 13 gives $a_i = a_{i+1}$ with the pair at an endpoint, that is $e(a) \geq 1$, and $\theta = \pi/4$. If the pair is interior with flanking legs $x = a_i \neq a_{i+2} = z$, then (4) has the rational root $u = a_{i+1}^2$, so its discriminant is a rational square and (x, z) is a nontrivial point of one of $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$, contradicting Lemma 14. If the pair is interior with $a_i = a_{i+2}$, Lemma 15 gives $a_i = a_{i+1} = a_{i+2}$, that is $t(a) \geq 1$, and $\theta = \pi/3$. This proves (iii) and the "only if" direction.

Conversely, if $t(a) \geq 1$ then some interior pair has $\theta = \pi/3$ by Lemma 15, the dihedral image has order 6, and the shortest relator is $(R_iR_{i+1})^3$ of length 6: (i). If $t(a) = 0$ and $e(a) \geq 1$ then some boundary pair has $\theta = \pi/4$ by Lemma 13, the dihedral image has order 8, and the shortest relator in that parabolic is $(R_iR_{i+1})^4$ of length 8; by the previous paragraph no other consecutive pair contributes an element of $\ker\rho_a$, since a boundary pair would force $e \geq 1$ with $\theta = \pi/4$ again and an interior pair would force $t \geq 1$: (ii). □

Following the terminology of the three-dimensional analysis, call a leg tuple *Class C* when its coordinates are pairwise distinct. Class C tuples have $e = t = 0$, so Theorem 16(iii) applies, but so do tuples with an interior equal pair such as $(1,2,2,3)$ and tuples with a non-adjacent repetition such as $(1,2,1,3)$; the latter is precisely the case that Lemma 15, and not Lemma 14, has to handle.

Conjecture 17 (Class C faithfulness)**.** *For every $n \geq 3$ and every $a \in \mathbb{Q}^n_{>0}$ with pairwise distinct coordinates, ρ_a is injective; equivalently $u_d(a) = [t^d]\, W_n(t)$ for every $d \geq 0$.*

Theorem 16(iii) is the rank-two half of Conjecture 17 and is proved. What is missing is that a nontrivial kernel element must lie in a rank-two parabolic, which is false in general: the planar case $n = 2$ is a counterexample. There the two consecutive pairs are both boundary pairs, so for $a_1 \neq a_2$ Theorem 16(iii) applies and no rank-two parabolic meets $\ker \rho_a$; yet $\ker \rho_a \neq \{1\}$ by Proposition 24, and its shortest elements have length 20, arising as $w_1 w_2^{-1}$ from eight pairs of distinct geodesic words w_1, w_2 of length 10 using all three generators [2]. Conjecture 17 is verified computationally at $n = 3$ with legs $(1,2,3)$ through depth 10 and at $n = 4$ with legs $(1,2,3,5)$ through depth 8 (§10).

5.5 A determinant identity, and what it does not give

The Lorentzian Gram matrix of the orthoscheme in staircase coordinates, with the standard $(n,1)$ endpoint correction, is the tridiagonal $(n+1) \times (n+1)$ matrix Q_n with

$$Q_n[0,0] = 1 - a_1^2, \qquad Q_n[i,i] = a_i^2 + a_{i+1}^2 \ (1 \leq i \leq n-1), \qquad Q_n[n,n] = 1,$$
$$Q_n[0,1] = a_2, \qquad Q_n[i,i+1] = a_i\, a_{i+2} \ (1 \leq i \leq n-2), \qquad Q_n[n-1,n] = a_{n-1}.$$

Lemma 18 (Uniform Schur-complement determinant identity)**.** *For every $n \geq 3$ and every positive tuple $(a_1, \ldots, a_n)$,*

$$\det Q_n = -\prod_{i=1}^{n} a_i^2.$$

Proof. Let $D_k = \det Q_n[:k{+}1, :k{+}1]$ be the leading principal minor of order $k+1$. Tridiagonal expansion along the last row gives

$$D_k = Q_n[k,k]\, D_{k-1} - Q_n[k-1,k]^2\, D_{k-2} \qquad (1 \leq k \leq n), \tag{†}$$

with $D_{-1} = 1$. We claim

$$D_k = \Big(\prod_{i=1}^{k} a_i^2\Big)\Big(1 - \sum_{i=1}^{k+1} a_i^2\Big) \qquad \text{for } 1 \leq k \leq n-1. \tag{⋆}$$

Base. $D_0 = 1 - a_1^2$ and $D_1 = (1-a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2)$.

Interior step. For $2 \leq k \leq n-1$ the recurrence (†) reads $D_k = (a_k^2 + a_{k+1}^2)D_{k-1} - (a_{k-1}a_{k+1})^2 D_{k-2}$. Introducing $S_{k-1} = \sum_{i=1}^{k-1} a_i^2$ and $P_{k-2} = \prod_{i=1}^{k-2} a_i^2$ and substituting (⋆) reduces the step to the polynomial identity

$$\begin{aligned} P_{k-2}\, a_{k-1}^2\, a_k^2\, (1 - S_{k-1} - a_k^2 - a_{k+1}^2) &= (a_k^2 + a_{k+1}^2)\, P_{k-2}\, a_{k-1}^2\, (1 - S_{k-1} - a_k^2) \\ &\quad - (a_{k-1}a_{k+1})^2\, P_{k-2}\, (1 - S_{k-1}) \end{aligned}$$

in the five independent symbols $(a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2})$. It is independent of k and of n.

Final step. At $k = n$ the recurrence uses $Q_n[n,n] = 1$ and $Q_n[n-1,n]^2 = a_{n-1}^2$, giving $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$. With $S_{n-1} = \sum_{i=1}^{n-1} a_i^2$ and $P_{n-2} = \prod_{i=1}^{n-2} a_i^2$,

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2}\, a_{n-1}^2 \big[(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})\big] = -P_{n-2}\, a_{n-1}^2 a_n^2 = -\prod_{i=1}^{n} a_i^2.$$

The three identities close the induction for every $n \geq 3$, with no per-n verification. All of them, the induction that assembles them, and the boundary step are machine-checked in Lean 4 over an arbitrary commutative ring; no positivity, ordering or characteristic hypothesis is used (§10). □

Remark 19 (Why nonvanishing of $\det Q_n$ does not give faithfulness). An earlier version of this material argued from Lemma 18 that $\det Q_n \neq 0$, hence the form is nondegenerate, hence ρ_a is faithful by Tits' theorem. That argument is a non sequitur, in three independent ways, and the deduction is withdrawn.

First, it proves too much. The hypothesis of Lemma 18 is that the legs are positive and nothing more, so the conclusion $\det Q_n \neq 0$ holds at every positive tuple, including $(1, 1, \ldots, 1)$. Were the inference valid, ρ_a would be faithful there too. It is not: breadth-first search at $n = 3$ with legs $(1, 1, 1)$ gives layer counts $1, 4, 9, 17, 28, 42$ against the envelope $1, 4, 9, 18, 36, 72$, a deficit already at depth 3, in agreement with Theorem 21, which assigns $\mathrm{cd}_3 = 3$ to that tuple. The same computation at $n = 4$ with legs $(1, 1, 1, 1)$ gives $1, 5, 14, 31, 59, 101$ against $1, 5, 14, 33, 75, 169$.

Second, Tits' theorem concerns the canonical geometric representation of an abstract Coxeter system, on the space spanned by the simple roots with the form $B(\alpha_i, \alpha_j) = -\cos(\pi/m_{ij})$ [3]. It is not a statement about ρ_a, which is a different representation, on $\mathbb{R}^n$ rather than $\mathbb{R}^{n+1}$ and affine rather than linear. Nor is Q_n a Coxeter Gram matrix: its diagonal is not identically 1, and $Q_n[0, 0] = 1 - a_1^2$ is negative whenever $a_1 > 1$.

Third, run at $n = 2$ the argument returns a false answer. There $\det Q_2 \neq 0$ just as well, so it would give $u_d(a) = [t^d]W_2(t) = F_{d+3}$ for all d; in fact the two agree only up to $d = 9$ and differ at $d = 10$, where the orbit count is 225 against the envelope's 233, a deficit of 8 accounted for by eight coincidences between distinct words of length 10 [2].

6. The collision depth

Definition 20. For $a \in \mathbb{Q}^n_{>0}$ the *collision depth* is

$$\mathrm{cd}_n(a) = \inf\{d \geq 0 : u_d(a) < [t^d]\, W_n(t)\} \in \{1, 2, \ldots\} \cup \{\infty\}.$$

By Corollary 6, $\mathrm{cd}_n(a) = K(a)/2$, and by Lemma 7, $K(a) \geq 6$. Conjecture 17 says exactly that $\mathrm{cd}_n = \infty$ for every Class C tuple. The next theorem describes cd_n for every leg multiplicity pattern. Although the multiplicity partition of $(a_1, \ldots, a_n)$ has $p(n)$ possible shapes, the collision depth takes only three values, determined by the two local positional statistics (e, t).

Theorem 21 (Collision depth, all $n \geq 3$)**.** *Let $n \geq 3$ and $a \in \mathbb{Q}^n_{>0}$. Then $\mathrm{cd}_n(a) \geq 3$, and*

$$\mathrm{cd}_n(a) = \begin{cases} \infty & \text{if } e = t = 0 \text{ and } \rho_a \text{ is injective,} \\ 4 & \text{if } e \geq 1,\ t = 0 \text{ and } K(a) \neq 6, \\ 3 & \text{if } t \geq 1. \end{cases}$$

The third case is unconditional. In the second case $\mathrm{cd}_n(a) \in \{3, 4\}$ unconditionally, and $K(a) = 6$ would require an element of $\ker \rho_a$ of length 6 lying in no rank-two standard parabolic; no such element occurs in any tuple of Table 2. In the first case $e = t = 0$ includes every Class C tuple, so that branch is ∞ exactly when Conjecture 17 holds.

Proof. $\mathrm{cd}_n(a) = K(a)/2$ with $K(a)$ even and at least 6, by Corollary 6 and Lemma 7; this gives $\mathrm{cd}_n(a) \geq 3$ in all cases.

If $t \geq 1$, Theorem 16(i) exhibits an element of $\ker \rho_a$ of length 6, so $K(a) \leq 6$, hence $K(a) = 6$ and $\mathrm{cd}_n(a) = 3$.

If $e \geq 1$ and $t = 0$, Theorem 16(ii) exhibits an element of length 8, so $6 \leq K(a) \leq 8$ and $K(a)$ is even, giving $\mathrm{cd}_n(a) \in \{3, 4\}$. By Theorem 16(ii) no element of a rank-two standard parabolic lies in $\ker \rho_a$ with length below 8, so $K(a) = 6$ is possible only through an element of length 6 lying in no rank-two standard parabolic. Absent such an element, $K(a) = 8$ and $\mathrm{cd}_n(a) = 4$.

If $e = t = 0$ then $\mathrm{cd}_n(a) = \infty$ if and only if $\ker \rho_a = \{1\}$, by Corollary 5. □

Remark 22 (What the second case leaves open). The gap in the second case is genuine and is a special case of the general problem: the rank-two analysis of §5 controls elements of $\ker \rho_a$ that lie in a dihedral standard parabolic, and nothing in this paper controls the others. The gap is narrow: only the single value $K(a) = 6$ is at stake, and only for tuples with an endpoint equal pair and no triple.

Remark 23 (The run-length rule fails in $n \geq 4$). A natural but incorrect heuristic is to declare cd_n finite as soon as some consecutive equal pair appears, that is as soon as the run length $R = \max_i \#\{j : a_j = a_i\}$ is at least 2. At $n = 3$ this rule coincides with the (e, t) rule, because every consecutive equal pair in a three-term sequence is automatically at an endpoint. At $n \geq 4$ the two diverge: the orthoscheme $(1, 2, 2, 3)$ has an interior equal pair, so $R = 2$ but $(e, t) = (0, 0)$, and Theorem 21 places it in the first case; empirically its layer counts $1, 5, 14, 33, 75, 169, \ldots$ are identical to those of the Class C tuple $(1, 2, 3, 5)$ through every computed depth, falsifying the run-length rule.

n	leg sequence	(e,t)	cd_n	first six layer counts
3	$(2,3,5)$	$(0,0)$	∞	$1,4,9,18,36,72$
3	$(1,1,2)$	$(1,0)$	4	$1,4,9,18,35,67$
3	$(1,1,1)$	$(2,1)$	3	$1,4,9,17,28,42$
4	$(1,2,3,5)$	$(0,0)$	∞	$1,5,14,33,75,169$
4	$(1,2,2,3)$	$(0,0)$	∞	$1,5,14,33,75,169$
4	$(1,1,2,3)$	$(1,0)$	4	$1,5,14,33,74,163$
4	$(1,1,2,2)$	$(2,0)$	4	$1,5,14,33,73,157$
4	$(1,1,1,2)$	$(1,1)$	3	$1,5,14,32,67,135$
5	$(1,2,3,5,7)$	$(0,0)$	∞	$1,6,20,54,136,334$
5	$(1,2,2,3,5)$	$(0,0)$	∞	$1,6,20,54,136,334$
5	$(1,1,2,3,5)$	$(1,0)$	4	$1,6,20,54,135,327$
5	$(2,1,1,1,2)$	$(0,1)$	3	$1,6,20,53,128,297$
6	$(1,1,2,3,2,2)$	$(2,0)$	4	$1,7,27,82,224,581$

Table 2: A small atlas of representative orthoschemes in $n = 3, 4, 5, 6$ illustrating each attainable (e, t) class. The layer counts are exact-rational breadth-first search (§10); in every row the observed collision depth is the value predicted by Theorem 21, and the entries ∞ are observed agreement with the envelope as far as computed, not a proof. The two $n = 4$ rows with $(e, t) = (0, 0)$ have identical counts through every computed depth despite differing in run structure.

The rows with $\mathrm{cd}_n = \infty$ saturate the upper bound $[t^d]\, W_n(t)$ as far as they are computed, as Conjecture 17 predicts; for $n = 3$ this reproduces $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

7. The plane, and why its argument does not extend

Proposition 24 (The plane)**.** *For $n = 2$ and every $a \in U$, ρ_a is not injective, so alternative (ii) of Corollary 5 holds.*

Proof. $W(a) \leq \mathrm{Isom}(\mathbb{R}^2) = \mathbb{R}^2 \rtimes O(2)$. The group $O(2)$ contains the abelian $SO(2)$ with index 2, so $\mathrm{Isom}(\mathbb{R}^2)$ is virtually solvable as an abstract group, hence amenable, and so is every subgroup of it; in particular $W(a)$ is amenable. The right-angled Coxeter group W_2 is not amenable: its defining graph Γ_2 is not complete multipartite, since its complement P_3 is connected and is not a disjoint union of cliques, and a right-angled Coxeter group whose defining graph is not complete multipartite contains a free subgroup of rank 2 [9]. An amenable group is not isomorphic to a non-amenable one, so ρ_a is not injective. □

Corollary 25 (The planar horizon)**.** *For $n = 2$ and every shape admissible in the sense of [2], the collision depth is $\mathrm{cd}_2 = 10$, equivalently $K(a) = 20$: $u_d(a) = F_{d+3} = [t^d]W_2(t)$ for $1 \leq d \leq 9$ (while $u_0 = 1$ and $F_3 = 2$), and at $d = 10$ the count is 225 against the envelope's 233. Past that horizon the sequence is A396406, with growth rate $\beta_2 = 1.4916\ldots < \varphi = r_2$, and its generating function is transcendental over $\mathbb{Q}(x)$ [2]. This is the only dimension in which the deviation has been computed past its onset.*

The isoceles right triangle $a_1 = a_2$ is excluded from that statement, and must be: there $e(a) = 1$, so Theorem 16(ii) gives an element of $\ker \rho_a$ of length 8 and hence $\mathrm{cd}_2 \leq 4$.

Remark 26 (The amenability trap: why the planar argument stops at $n = 2$). The proof of Proposition 24 does not extend to $n \geq 3$, and the natural extension is false. One is tempted to argue that $\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$ is amenable because $O(n)$ is compact. Compactness gives amenability of $O(n)$ as a *topological* group, with respect to Haar measure, and that property does not pass to abstract subgroups. As an abstract discrete group $O(n)$ is non-amenable for every $n \geq 3$: already $SO(3)$ contains a free subgroup of rank 2 (Hausdorff; see [21, Thm. 2.1]), which is the standard ingredient of the Banach–Tarski paradox. Hence $\mathrm{Isom}(\mathbb{R}^n)$ is non-amenable as an abstract group for $n \geq 3$ and yields no obstruction at all. At $n = 2$ the argument survives only because $O(2)$ is virtually abelian.

8. What is proved, and the open problems

Table 3 lists the status of every numbered statement. Four things are proved unconditionally and dimension-freely: the dichotomy (Theorem 4); generic rigidity (Theorem 2), which makes "the generic n-orthoscheme growth" a well-defined object realised by a co-null set of shapes; rationality of the envelope with rate r_n (Theorem 8); and the rank-two classification (Theorem 16), with its consequences that $\mathrm{cd}_n \geq 3$ for every leg tuple and that $\mathrm{cd}_n = 3$ at every tuple with three consecutive equal legs. Whether $\mathrm{cd}_n = 3$ can also occur without such a triple is not decided here; it would require a kernel element of length 6 in no rank-two parabolic (Remark 22).

Which alternative of Corollary 5 holds is settled only in the plane, by Proposition 24, where it is (ii) with horizon 10. For $3 \leq n \leq 7$ the geometric growth has been computed to depth 13 to 16 and agrees with the envelope throughout that range. That agreement is consistent with both alternatives: under (i) it would continue forever, and under (ii) it would end at a horizon beyond the computed range. Nothing here decides between them.

Problem 27 (The alternative in dimension $n \geq 3$)**.** *Fix $n \geq 3$.*

(a) Generic form. *Is $N_{\mathrm{gen}} = \bigcap_{a \in U} \ker \rho_a$ trivial? Equivalently, by Theorem 2, is ρ_a injective for every a in the co-null set U_{gen}; equivalently, by Corollary 5, is the growth series of the generic n-orthoscheme group the rational envelope $W_n(t)$ in every degree?*

(b) Arithmetic form. *Is ρ_a injective for every Class C tuple $a \in \mathbb{Q}^n_{>0}$? This is Conjecture 17.*

The two forms are related in one direction only. By Corollary 3, a single faithful tuple, rational or not, answers (a) affirmatively; so (b) implies (a). The converse fails to be automatic because U_{gen} contains no tuple with algebraically dependent coordinates, hence no rational tuple: an affirmative answer to (a) says nothing directly about any given rational a. If the answer to (a) is affirmative for all $n \geq 3$, the plane is the unique dimension with non-rational orbit growth. If instead $\ker \rho_a$ is nontrivial in some dimension $n \geq 3$, the growth deviates below the envelope at $\mathrm{cd}_n(a) = K(a)/2$, and the arithmetic of the deviated series past that horizon is open even in the cases where the horizon can be located.

Three obstructions are worth stating precisely, because they close off the routes that look available.

(O1) *The rank-two half is all that the Diophantine method gives.* Faithfulness would follow from Theorem 16(iii) if every nontrivial kernel element lay in a rank-two parabolic. That implication is false in general: at $n = 2$ with $a_1 \neq a_2$ no rank-two parabolic meets $\ker \rho_a$, yet $\ker \rho_a$ is nontrivial, its shortest elements having length 20 and using all three generators [2]. So no soft upgrade of Theorem 16 to Conjecture 17 is available, and the same obstruction is what leaves the value $K(a) = 6$ unexcluded in the second case of Theorem 21 (Remark 22).

(O2) *The amenability trap.* The one soft argument that settles $n = 2$ is unavailable for $n \geq 3$, and not merely unproved: $O(n)$ contains a free subgroup of rank 2 as an abstract group for every $n \geq 3$, so $\mathrm{Isom}(\mathbb{R}^n)$ is not amenable as an abstract group and imposes no constraint whatsoever on its subgroups' isomorphism type (Remark 26).

(O3) *Finite presentation.* At $n = 2$, Isola and Piergallini [13] show that the generic triangle reflection group is not finitely presented, which gives non-injectivity independently of (O2). We do not know whether the generic n-orthoscheme reflection group is finitely presented for $n \geq 3$; an answer either way would decide Problem 27(a), since W_n is finitely presented.

Three further questions are open and are stated for the record.

(Q1) *Other diagrams.* Does an analogue of Theorem 21 hold for other families of simplicial reflection chambers in $\mathbb{R}^n$, for instance Coxeter orthoschemes with bent paths or simplices from non-path-shaped diagrams? The analysis here uses the right-angled-with-one-nontrivial-bond structure of the path diagram throughout.

(Q2) *Intermediate rates.* In every dimension $n \geq 3$, multiplicity patterns that are neither Class C nor saturated by triples produce growth rates strictly between two consecutive values of r_n. Whether these intermediate rates are algebraic over $\mathbb{Q}$ at all is open.

(Q3) *Beyond (e, t).* Theorem 21 shows that the collision depth is determined by the positional statistics (e, t). Is there a finer positional invariant that determines the entire sequence $u_d(a)$, not only its first deviation?

9. Related work

Path Coxeter groups and growth-rate calculus. The closed form $r_n = 1 + 2\cos(2\pi/(n + 3))$ is the signature of a linear, path-shaped Coxeter group whose Poincaré-series denominator reduces to a tridiagonal Chebyshev-type characteristic polynomial. Cannon and Wagreich [4], Floyd and Plotnick [11] and Parry [19] developed the growth-rate calculus of such groups, and Kellerhals and Perren [14] treat right-angled Coxeter growth systematically. Theorem 8 fits into that framework with parameter $m = n + 3$.

A Salem-type family on the surface-dynamics side. The family $\{1 + 2\cos(2\pi/m)\}_{m \geq 3}$ to which r_n belongs also arises in birational surface dynamics. Diller and Favre [10] introduced the dynamical-degree framework for bimeromorphic maps of surfaces; McMullen [17] showed that this family arises as dynamical degrees of automorphisms of blowups of $\mathbb{P}^2$ associated with

Statement	Status	Note
Lemma 1 (facet normals)	proved (Lean)	`OrthoschemeNormals`
Theorem 2 (generic rigidity)	proved	
Corollary 3	proved	
Theorem 4 (dichotomy)	proved (Lean)	`EnvelopeDichotomy`
Corollaries 5, 6	proved	
Lemma 7 ($K \geq 6$)	proved	
Theorem 8 (envelope)	proved (Lean, partly)	`EnvelopeDichotomy`, `NdimRate`
Corollary 11	proved (Lean)	`NdimRate`
Lemma 13 (boundary pairs)	proved (Lean)	`RankTwoExclusion`
Lemma 14 (three quartics)	proved	rank-0 descents, not formalised
Lemma 15 (equal flanking legs)	proved	
Theorem 16 (rank-two classification)	proved	interior half uses Lemma 14
Lemma 18 ($\det Q_n$)	proved (Lean)	`OrthoschemeDet`
Theorem 21, case $t \geq 1$	proved	
Theorem 21, case $e \geq 1, t = 0$	$\mathrm{cd}_n \in \{3, 4\}$ proved	value 4 conditional, Rem. 22
Theorem 21, case $e = t = 0$	conditional	equivalent to Conj. 17
Conjecture 17 (Class C)	conjecture	checked to depth 10 ($n = 3$), 8 ($n = 4$)
Proposition 24 (the plane)	proved	
Problem 27	open in both directions	

Table 3: Status of the numbered statements. "Lean" means that the indicated Lean 4 file, which builds against Mathlib with no `sorry`, covers the statement; "partly" means that it covers some steps of the proof and not others. The scope of each file is described in §10.

the E_n T-shaped Coxeter graph, which is not the path graph occurring here; and Bedford and Kim [1] realised specific members through linear-fractional recurrences. The rate r_n lies in this family, but the right-corner orthoscheme reflection group is not known to be realised as a McMullen involution product on a rational surface, and the coincidence is at the level of the algebraic-number family only. A birational-geometry construction realising r_n as a dynamical degree on a $\mathbb{P}^2$-blowup would be of independent interest.

10. Certification

Orbit counts. The layer counts quoted above, the envelope coefficients and the Diophantine searches are produced by `code/zeta_probe/tools/ortho_cd` (Rust, exact arithmetic in $\mathbb{F}_p$ at two primes). It builds the $n+1$ facet reflections directly from the legs, enumerates the group by breadth-first search, and compares layer by layer against $(1+t)^{n+1}/J_{n+1}(t)$. In particular it produces the counts of Remark 19 at $(1,1,1)$ and $(1,1,1,1)$ and at $n = 2$, confirms the (e,t) prediction on every tuple of Table 2 including the non-adjacent repetition $(1,2,1,3)$, and searches the three quartics of Lemma 14 for nontrivial solutions with legs up to 300, finding none, while exhibiting rational points of the corresponding conics in (X,Z) to show that the restriction to squares is not cosmetic.

The envelope. Theorem 8 is certified by `code/zeta_probe/tools/racg_envelope` (Rust, exact integer polynomial arithmetic), which for $2 \leq n \leq 30$ checks that the recurrence for J_m and the Steinberg expansion agree coefficient by coefficient, performs the division by $(1+t)^{\lfloor (n+1)/2 \rfloor}$ exactly, and matches the smallest positive root of K_{n+1} against $1/\big(1+2\cos\frac{2\pi}{n+3}\big)$. Independently, for $2 \leq n \leq 6$ it enumerates W_n by breadth-first search in the canonical geometric representation, whose matrices are integral here because the bilinear form takes only the values $1, 0, -1$ and which is faithful by Tits' theorem [3]; the sphere sizes agree with the series coefficients to depths 9 to 14. The same program records two checks on the shape of the answer: the variant substitution

$-1/(1+t)$ in place of $-t/(1+t)$ gives a different rational function already at $n = 2$, and the polynomial $1 - 2t - 3t^2 + 4t^3 - t^4$ is not the $n = 7$ denominator, being the negated reciprocal of $1 - 4t + 3t^2 + 2t^3 - t^4$ and predicting 6 elements at depth 1 in a group with 8 generators.

Lean 4. Five files in `lean/with_mathlib/` build against Mathlib with no `sorry`.

Paper result	Lean declaration
OrthoschemeNormals.lean (7 theorems)	
Lemma 1, support pattern	`supp_disjoint`
Lemma 1, orthogonality on disjoint supports	`mul_eq_zero_of_disjoint`, `sum_eq_zero_of_disjoint`
Lemma 1, the three adjacent products	`dot_zero_one`, `dot_interior`, `dot_last`
Lemma 1, combined	`orthogonality_pattern`
EnvelopeDichotomy.lean (12 theorems)	
Thm. 4, horizon arithmetic	`two_mul_lt_of_lt_horizon`, `split_lengths`
Thm. 4, even case	`card_image_lt_of_collision`
Thm. 4, odd case	`card_lt_of_escapes`
Thm. 8, three-term recurrence	`cos_three_term`, `sin_three_term`, `J_recurrence`
Thm. 8, root condition	`cot_add_cot_eq_zero_iff`, `J_eq_zero_iff`
Thm. 8, recovering t from θ	`t_of_theta`, `rate_of_theta`
Remark 9	`rate_ne_path_spectral_radius`
NdimRate.lean (7 theorems)	
Remark 10, Pascal step	`pascal_step` (no axioms), `coeff_pascal`
Remark 10, Binet formula	`sq_eq`, `pell_binet`
Remark 10, root translation	`root_translation`
Corollary 11	`rate_residual`, `three_sub_r`
RankTwoExclusion.lean (10 theorems)	
Lemma 13, the five Niven values	`niven_zero`, `niven_one`, `niven_quarter`, `niven_three_quarter`, `niven_half`, `boundary_no_relation`
Lemma 13, 3 is not a rational square	`sq_ne_three`, `not_isSquare_three_int`
Boundary cosine nonzero, and $\cos^2 \neq \frac{1}{4}$ there	`no_length_two`, `no_length_three_boundary`
OrthoschemeDet.lean (3 theorems)	
Lemma 18, closed form ($\star$)	`minor_closed_form`
Lemma 18, boundary step	`det_Q`
Lemma 18, full statement	`det_Q_of_recurrence`

`NdimRate.lean` and `OrthoschemeDet.lean` end with `#print axioms`. Of the ten declarations they contribute to the table, six report exactly the standard axioms `propext`, `Classical.choice`, `Quot.sound`, and four report strictly fewer, with `pascal_step` depending on no axioms at all.

What is not formalised. `RankTwoExclusion.lean` covers the boundary half of Theorem 16 only; the interior cases rest on the rank-0 Mordell descents of Lemma 14 and are not formalised. `EnvelopeDichotomy.lean` formalises the arithmetic of Theorem 4 and the trigonometry of the root identity, but no word metric, no Coxeter group and no clique polynomial: the group theory around those steps is on paper. The step identifying the growth rate with the largest root t_1 is a dominant-pole argument on a rational generating function, proved on paper; so is the induction assembling the Pascal step into $c_{\Gamma_n} = P_{n+3}$ in Remark 10. Everything downstream of those two inputs is machine-checked. Lemma 7, Lemma 15 and Theorem 2 are paper proofs.

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